1.3.1 Properties of Subtraction

We can test whether all the properties of integers that are true for addition still hold for subtraction or not.

Recall that subtraction of two whole numbers did not always result in a whole number. However, this extended form of integers is enough to make sure that the difference of two integers is also an integer. For example, $(-7)-(-2)$ is an integer, $(-5)+14$ is an integer and $0-(-8)$ is an integer. From these examples we observe that the collection of integers is "closed", that is, the difference of two integers is always an integer.

Therefore, for any two integers a, b ; $a-b$ is also an integer.

What about the other properties? Can you see that $(-2)-(-5)=3$ but $(-5)-(-2)=-3$ Also, $10-(-5)=15$ but $(-5)-10=-15$. Therefore, changing the order of integers in subtraction will not give the same value. Hence, the commutative property does not hold for subtraction of integers.

Therefore, for any two integers a, b ; $a-b \neq b-a$.

Try these

1. Fill in the blanks.
   (i) $(-7)-(-15)=$____ (ii) $12-$___ $=19$ (iii) ___ $-(-5)=1$
2. Find the values and compare the answers.
   (i) 15 – 12 and 12 –15 (ii) –21 –32 and –32 –(–21)

Think

Is assosiative property true for subtraction of integers?
Take any three examples and check.

Exercise 1.2

1. Fill in the blanks
   (i) $-44+$___ $=-88$ (ii) ____ $-75=-45$ (iii) ____ $-(+50)=-80$
2. Say true or false
   (i) $(-675)-(-400)=-1075$
   (ii) $15-(-18)$ is the same as $15+18$
   (iii) $(-45)-(-8)=(-8)-(-45)$
3. Find the value of the following
   (i) –3–(–4) using number line (ii) 7–(–10) using number line
   (iii) 35 – (–64) (iv) –200 – (+100)
4. Kabilan was having 10 pencils with him. He gave 2 pencils to Senthil and 3 to Karthik. Next day his father gave him 6 more pencils, from that he gave 8 to his sister. How many pencils are left with him?

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